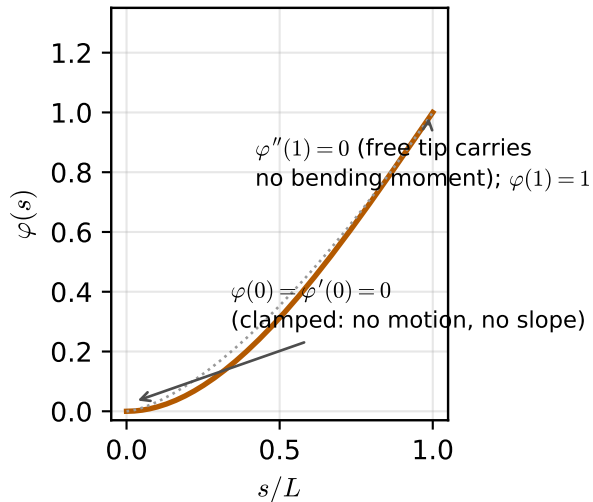
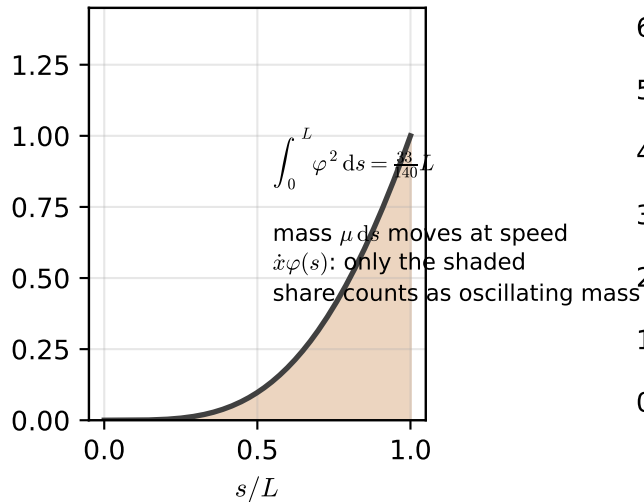


deflection shape $y(s, t) = x(t) \varphi(s)$



where the moving mass is ($\propto \varphi^2$)



where the bending is ($\propto \varphi''^2$)

